CIS-OpenSLU at NADI 2026 Shared Task: ASR-Matched Training and Confidence-Based Rejection for Open-Set Tunisian Arabic Intent Recognition

Salma Khaled Hegazy
Center for Informatics Science (CIS), School of Information Technology and Computer Science
Nile University, Giza 12588, Egypt
SaHegazy@nu.edu.eg

Abstract

Open-set spoken intent recognition for low-resource Arabic varieties must handle both automatic speech recognition (ASR) errors and intents absent from training. We describe CIS-OpenSLU, our NADI 2026 Task 5.1 system, which transcribes Tunisian Arabic speech with Whisper Turbo and classifies the resulting text using word- and character-level TF–IDF features with a LinearSVC. On ASR-transcribed validation audio, training the classifier on Whisper-generated transcripts rather than gold transcripts raises weighted F1 from 49.25% to 69.78%, showing that matching training and inference text distributions is crucial. We rank test utterances by the maximum SVM decision score and reject low-confidence examples as `unknown`; the final system achieves 71.54% weighted F1 and 72.50% accuracy, ranking second. Auxiliary audio and multimodal experiments further show that strong unseen-intent detection in leave-one-intent-out simulation does not necessarily transfer to genuinely unseen test intents.

1 Introduction

Spoken language understanding (SLU) maps speech to semantic representations that support applications such as task-oriented dialogue and voice assistants. Intent recognition is one of its central components: given an utterance, a system must determine the action or communicative goal expressed by the speaker. Although multilingual speech and language models have improved SLU substantially, dialectal Arabic remains challenging because labeled resources are limited and spoken varieties exhibit considerable phonetic, lexical, and orthographic variation.

Tunisian Arabic introduces additional difficulties through spelling variation, code-switching, and the mismatch between written conventions and spontaneous speech. In a cascaded SLU system, these difficulties are amplified by ASR errors that propagate to downstream intent classification. NADI 2026 Task 5.1 (Sullivan et al., 2026) evaluates this setting using Tunisian Arabic speech and an open-set protocol in which the blind test data can contain intents absent from the released training inventory.

We address the task using a lightweight cascaded pipeline. First, Whisper Turbo (Radford et al., 2023) transcribes the training, validation, and blind-test recordings. Second, a linear support vector classifier predicts the intent from a union of word- and character-level TF–IDF features. Rather than training the downstream classifier only on human-written transcripts, we train it on Whisper-generated transcripts, exposing it to the same type of recognition noise encountered during inference.

To handle unseen intents, we use the maximum SVM decision score as a confidence signal. Low-confidence examples are mapped to the special `unknown` label. We additionally investigate frozen XLS-R acoustic representations (Babu et al., 2022), late audio–text fusion, and a learned multimodal unknown-intent router.

Our main contributions are: (i) an ASR-matched training strategy that substantially reduces the mismatch between clean training text and noisy test-time transcripts; (ii) a computationally efficient word–character TF–IDF and LinearSVC classifier with confidence-based open-set rejection; (iii) a controlled comparison of transcript-based, direct-audio, and multimodal approaches; and (iv) evidence that leave-one-intent-out open-set simulation may not reliably predict performance on genuinely unseen test intents.

The final system achieves 71.54% weighted F1 and 72.50% accuracy, ranking second in NADI 2026 Task 5.1. More broadly, the work supports inclusive Arabic NLP by studying a resource-efficient SLU approach for an underrepresented Arabic variety.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 72–78
October 28-29, 2026 ©2026 Association for Computational Linguistics

2 Related Work

The NADI shared-task series initially focused on nuanced Arabic dialect identification from 2020 to 2024 (Abdul-Mageed et al., 2020, 2021, 2022, 2023, 2024), before expanding to multidialectal Arabic speech processing in NADI 2025 (Talafha et al., 2025). NADI 2026 continues this speech-processing direction (Sullivan et al., 2026).

SLURP provides a benchmark for task-oriented spoken language understanding (Bastianelli et al., 2020), while SLURP-TN extends this setting to Tunisian Arabic (Elleuch et al., 2026). For speech modeling, we use Whisper (Radford et al., 2023) as a frozen ASR front end and investigate frozen XLS-R representations (Babu et al., 2022) for direct-audio classification. Open-set intent recognition has previously been studied through out-of-scope prediction (Larson et al., 2019) and out-of-domain detection (Shen et al., 2021). Our setting combines these challenges with ASR noise and unseen intents in dialectal Arabic.

3 Task and Data

NADI 2026 Task 5.1 (Sullivan et al., 2026) evaluates spoken intent recognition for Tunisian Arabic. Given an audio recording x, a system predicts an intent label y. The evaluation is open-set: some test utterances correspond to intents absent from the released training inventory and can therefore be assigned the special `unknown` label. For a test-only reference intent, the official scorer awards credit to either `unknown` or the exact fine-grained test label. Since our final classifier contains only released train+validation labels, it can obtain credit on test-only intents only through `unknown`.

We use the official SLURP-TN resource (Elleuch et al., 2026). The training split contains 2,677 utterances and validation contains 595. A labeled research test split contains 893 utterances; our inspection was limited to checking schema, split sizes, and label inventory, and it was not used for model fitting, hyperparameter or threshold selection, or the reported validation metrics. The official blind test contains 989 audio recordings with identifiers but no released transcriptions or labels.

The training split contains 21 intent labels. Two additional labels, `Emails` and `set`, occur only in validation. Full-validation results in Tables 2 and 3 retain these examples under their original labels: they are neither excluded nor mapped to `unknown`. Therefore, classifiers trained only on the training split have 21 output classes and cannot correctly predict those two validation-only labels. Seen-only diagnostics were computed separately during development but are not the main results reported here.

Split	Utterances	Labels	Use
Train	2,677	Yes	Development
Validation	595	Yes	Local evaluation
Research test	893	Yes	Inspection only
Blind test	989	No	Official submission

Table 1: Data splits used in this work.

After model selection, the final blind-test text classifier is retrained on the 2,677 training and 595 validation ASR transcripts, yielding 3,272 examples and 23 released labels. This same final closed-set model supplies the known-intent predictions and text confidence scores used by all text-confidence rows in Table 5. We use no external labeled intent-recognition data and no synthetic text or speech augmentation.

4 System Description

4.1 Whisper Turbo ASR

We use Whisper Turbo through the `faster-whisper turbo` alias (large-v3-turbo). Whisper is frozen and is not fine-tuned on SLURP-TN. No model revision was explicitly pinned. The execution notebook installed `faster-whisper` with an unpinned upgrade command, so the exact library revision cannot be recovered from the exported notebook; we report this as a reproducibility limitation rather than infer a version retrospectively.

Decoding uses `language=ar`, `task=transcribe`, beam size 5, temperature 0, `condition_on_previous_text=False`, and `vad_filter=False`. Inference uses FP16 on an NVIDIA T4 GPU, with INT8–FP16 fallback, and required roughly 27, 6, and 10 minutes for training, validation, and blind-test audio, respectively. Decoded segments are concatenated and only whitespace is normalized; scripts, punctuation, numerals, code-switching, and spelling variation are preserved. TF–IDF extraction and LinearSVC training run on CPU.

4.2 ASR-Matched Training

A conventional cascade can train an intent classifier on human transcripts and apply it to ASR text

at test time, creating a train–inference mismatch. We instead transcribe the labeled training audio with the same Whisper configuration used at inference and train the classifier on those generated transcripts. This changes the downstream training distribution without updating Whisper parameters and is therefore not ASR fine-tuning.

The Gold+ASR condition in Table 2 treats the two textual views as separate training instances: each of the 2,677 training recordings contributes its gold transcript and its Whisper transcript with the same intent label, for 5,354 text instances.

4.3 Word and Character TF–IDF

Each transcript is represented by a feature union of word- and character-level TF–IDF vectors. The word component uses 1–2 grams, `min_df=1`, `max_df=0.995`, sublinear term frequency, and at most 60,000 features. The character component uses `char_wb` 2–5 grams, `min_df=2`, sublinear term frequency, and at most 120,000 features. All unspecified `TfidfVectorizer` options used the execution-environment defaults, including L2 normalization, Latin-script lowercasing, and the default word token pattern. Character features provide partial matching under spelling variation, transliteration, code-switching, and ASR errors.

4.4 LinearSVC Intent Classifier

The concatenated sparse representation is classified using scikit-learn's LinearSVC (Pedregosa et al., 2011). We use $C = 2.0$, no class weighting, and random seed 42. The text value $C = 2.0$ was a fixed design choice rather than the result of an exhaustive hyperparameter search. Given decision scores $s_k(x)$ for known intents k,

$$\hat{y}_{\text{closed}}(x) = \arg\max_k s_k(x), \tag{1}$$

$$c(x) = \max_k s_k(x). \tag{2}$$

The maximum score $c(x)$ is used as a confidence signal.

4.5 Rank-Based Open-Set Rejection

SVM decision values are margins rather than calibrated probabilities, so we use their relative ordering. The 989 blind-test examples are ranked by $c(x)$ from low to high, and the final system maps the first 740 to `unknown`; the remainder retain their closed-set prediction. Ties are handled by a stable sort, preserving original blind-test order:

$$\hat{y}(x_i) = \begin{cases} \textsf{unknown}, & \text{rank}_{\uparrow}(c(x_i)) \le 740, \\ \hat{y}_{\text{closed}}(x_i), & \text{otherwise.} \end{cases} \tag{3}$$

The 740 count was refined through the permitted test-phase leaderboard and is therefore test-informed. None of the rows in Table 5 is a rejection operating point selected solely on development data. We did not calibrate per-class SVM margins or explicitly verify margin comparability across predicted classes; this is one reason to interpret the method as batch ranking rather than a calibrated probability of novelty.

4.6 Audio, Fusion, and Simulated Unknowns

For direct audio classification, a frozen `torchaudio.pipelines.WAV2VEC2_XLSR_300M` encoder produces 1,024-dimensional frame representations. Masked mean and standard-deviation pooling yields 2,048-dimensional utterance vectors, which are standardized and classified by LinearSVC. We search $C \in \{0.0003, 0.001, 0.003, 0.01, 0.03, 0.1\}$ and select $C = 0.001$ on validation. The notebook enforces matching PyTorch and TorchAudio release versions at runtime, but the printed version strings were not preserved in the exported notebook.

For late fusion, text and audio class-score vectors are aligned and z-normalized independently for each utterance, $z_{ik} = (s_{ik} - \mu_i)/\max(\sigma_i, 10^{-6})$. We search audio weights from 0 to 0.40 in increments of 0.025; the best validation mixture uses 95% text and 5% audio.

For open-set simulation, we select 14 training intents with at least 20 examples. In each leave-one-intent-out run, all examples of one intent become simulated unknowns. The remaining intents are split into approximately 80% known training and 20% known evaluation examples (singleton known classes remain in training). Fresh text TF–IDF/LinearSVC and audio LinearSVC models are then fit only on the known-training subset and evaluated on the known evaluation examples plus all examples of the held-out intent.

The multimodal router uses 10 features: text/audio top scores, top-vs-second score gaps, row-normalized top scores and gaps, modality agreement, and the absolute difference between normalized top scores. A StandardScaler + logistic regression model uses $C = 1.0$, balanced class

weights, seed 42, and `max_iter=5000`. Router probabilities are evaluated out of fold with seven-fold GroupKFold grouped by held-out intent. This grouping prevents a simulated held-out-intent run from appearing in both router training and evaluation folds; however, the same underlying utterance can appear as a known example in different simulations, so these router metrics are diagnostic rather than a fully nested deployment estimate.

5 Results

We report accuracy and weighted F1 for the local intent-classification experiments. Weighted F1 is the primary shared-task metric.

5.1 Effect of ASR-Matched Training

Applying the gold-trained model to Whisper transcripts reduces weighted F1 from 78.74% to 49.25%. Training on Whisper-generated transcripts restores it to 69.78%, a gain of **20.53 percentage points** over Gold→ASR. Duplicating each training recording with both gold and ASR text does not improve performance (68.76% versus 69.78%), suggesting that adding clean-text views does not resolve the inference-time distribution mismatch.

5.2 Direct Audio and Late Fusion

The frozen XLS-R system is substantially weaker than the ASR-text classifier. This likely reflects a combination of frozen representations, simple order-insensitive pooling, and limited labeled data rather than a general limitation of audio-based intent models. Late fusion yields only a 0.31-point weighted-F1 gain, so the final submitted classifier remains text based.

5.3 Open-Set Simulation

The multimodal router is substantially stronger under the simulated protocol. This advantage does not carry over to the official test submissions: with both methods constrained to reject exactly 590 examples, text-confidence ranking obtains 69.63% weighted F1, whereas multimodal ranking obtains 60.15%. Aggregate blind-test scores do not identify which error type caused the loss, so we treat mismatch between simulated and true unseen intents as a plausible explanation rather than a demonstrated causal mechanism.

5.4 Official Test-Phase Results

The final system achieves 71.54% weighted F1 and 72.50% accuracy, ranking second. Post-hoc official diagnostics, unavailable during system selection, identify 349 seen-intent utterances and 640 test-only utterances in the 989-example blind batch. Thus, test-only intents constitute 64.71% of the batch, while the final system rejects 740/989 = 74.82%. The high optimal rejection rate is therefore partly explained by the unusual test composition, but the final count should not be interpreted as a transferable deployment prior.

6 Discussion

The strongest controllable improvement comes from matching the downstream classifier to the ASR output distribution. A classifier trained only on clean human transcripts performs poorly when it receives Whisper text, while training directly on Whisper-generated transcripts recovers much of the lost performance. The Gold+ASR result further suggests that simply duplicating examples with clean and noisy textual views is less useful than matching the classifier specifically to its inference distribution.

The direct-audio XLS-R system underperforms because its frozen representations are not explicitly optimized for intent semantics and mean–standard-deviation pooling discards temporal ordering. The small fusion gain indicates limited complementarity but does not justify the additional complexity in the final system.

The multimodal router provides a useful negative result. Its strong leave-one-intent-out diagnostics do not transfer to the blind-test aggregate score. One plausible explanation is that artificially held-out intents remain too similar to known data in speakers, recording conditions, annotation conventions, vocabulary, and ASR behavior. The protocol also reuses some utterances as known examples across different simulated holdouts. We therefore interpret the router experiment as evidence that proxy-unknown validation can be misleading, not as proof that the blind-test degradation is specifically an unknown-detection failure.

For a new unlabeled batch, the fixed count of 740 should not be reused. A practical system would need representative held-out data containing known and proxy-unknown intents, or a better calibrated novelty score, to choose a threshold or rejection rate. Our results show that the leave-one-intent-out proxy used here was insufficiently predictive, so robust operating-point selection remains an open problem.

System	Training input	Evaluation input	Classifier	Accuracy	Weighted F1
Majority	—	—	Majority label	20.67	7.08
Gold → Gold	Gold	Gold	TF–IDF + LinearSVC	78.99	78.74
Gold → ASR	Gold	Whisper	TF–IDF + LinearSVC	49.08	49.25
ASR → ASR	Whisper	Whisper	TF–IDF + LinearSVC	**70.25**	**69.78**
Gold+ASR → ASR	Gold + Whisper	Whisper	TF–IDF + LinearSVC	69.08	68.76

Table 2: Local validation results. The arrow denotes training transcript source → evaluation transcript source. Values are percentages; boldface marks the best system among rows evaluated on ASR input.

System	Accuracy	Weighted F1
ASR text	70.25	69.78
Frozen XLS-R audio	28.07	27.45
Text–audio fusion	**70.59**	**70.09**

Table 3: Validation comparison of text, audio, and late fusion. Values are percentages.

Detector	AUC	AP	U-Recall	U-F1
Text confidence	.695	.399	.376	.395
Multimodal router	**.810**	**.569**	**.645**	**.594**

Table 4: Leave-one-intent-out diagnostics. Recall and F1 are for the simulated `unknown` class at approximately 20% known-intent rejection. Router probabilities are seven-fold out-of-fold predictions grouped by held-out intent.

System	Rejected (n)	Accuracy	W-F1
Text confidence	344	52.38	57.54
Text confidence	497	61.58	65.49
Text confidence	590	67.14	69.63
Multimodal router	590	57.84	60.15
Text confidence	700	70.68	70.80
Final system	740	**72.50**	**71.54**

Table 5: Six successive official test-phase submissions. Text-confidence rows share the same 23-class closed-set model and differ only in rejection count; the multimodal row changes the ranking used to select rejected examples. Values are percentages.

7 Conclusion

We presented CIS-OpenSLU for NADI 2026 Task 5.1, combining Whisper Turbo, word- and character-level TF–IDF, LinearSVC, and confidence-based rejection. ASR-matched training improved robustness to transcription noise, while XLS-R and multimodal extensions offered limited or non-transferable gains. The final system ranked second with 71.54% weighted F1 and 72.50% accuracy. The results underscore the importance of train–inference matching and realistic calibration of open-set evaluation for Arabic SLU.

Limitations

Whisper Turbo is used without Tunisian-specific fine-tuning, so ASR errors remain an important source of downstream mistakes. The final classifier depends on ASR text, and a recognition error can remove a decisive lexical cue in a short command.

LinearSVC decision scores are margins rather than calibrated probabilities. We do not calibrate scores across classes or verify class-wise margin comparability, and the final rejection count is test-informed and batch-specific. The leave-one-intent-out simulation also does not reproduce the genuine unseen-intent distribution and is not fully nested at the utterance level.

The direct-audio experiment uses frozen XLS-R features with simple statistical pooling; its weak result should therefore not be interpreted as evidence that end-to-end or fine-tuned acoustic intent models are unsuitable. The supervised dataset is relatively small, particularly for rare classes, and two labels are absent from the training split.

Finally, the execution notebook used unpinned upgrades for several Python packages, so exact `faster-whisper`, scikit-learn, PyTorch, TorchAudio, and CTranslate2 versions cannot all be reconstructed from the exported code. Future releases should pin package and model revisions. The system also maps test-only intents to a generic `unknown` label and is evaluated on one Tunisian Arabic benchmark, limiting claims about other dialects, domains, and intent inventories.

References

Muhammad Abdul-Mageed, Chiyu Zhang, Houda Bouamor, and Nizar Habash. NADI 2020: The first nuanced Arabic dialect identification shared task. In *Proceedings of the Fifth Arabic Natural Language Processing Workshop*, pages 97–110, Barcelona, Spain (Online), December 2020. Association for Computational Linguis-

tics. URL https://aclanthology.org/2020.wanlp-1.9.

Muhammad Abdul-Mageed, Chiyu Zhang, Abdel-Rahim Elmadany, Houda Bouamor, and Nizar Habash. NADI 2021: The second nuanced Arabic dialect identification shared task. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 244–259, Kyiv, Ukraine (Virtual), April 2021. Association for Computational Linguistics. URL https://aclanthology.org/2021.wanlp-1.28.

Muhammad Abdul-Mageed, Chiyu Zhang, Abdel-Rahim Elmadany, Houda Bouamor, and Nizar Habash. NADI 2022: The third nuanced Arabic dialect identification shared task. In *Proceedings of the The Seventh Arabic Natural Language Processing Workshop (WANLP)*, pages 85–97, Abu Dhabi, United Arab Emirates (Hybrid), December 2022. Association for Computational Linguistics. URL https://aclanthology.org/2022.wanlp-1.9.

Muhammad Abdul-Mageed, AbdelRahim Elmadany, Chiyu Zhang, El Moatez Billah Nagoudi, Houda Bouamor, and Nizar Habash. NADI 2023: The fourth nuanced Arabic dialect identification shared task. In Hassan Sawaf, Samhaa El-Beltagy, Wajdi Zaghouani, Walid Magdy, Ahmed Abdelali, Nadi Tomeh, Ibrahim Abu Farha, Nizar Habash, Salam Khalifa, Amr Keleg, Hatem Haddad, Imed Zitouni, Khalil Mrini, and Rawan Almatham, editors, *Proceedings of ArabicNLP 2023*, pages 600–613, Singapore (Hybrid), December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.arabicnlp-1.62. URL https://aclanthology.org/2023.arabicnlp-1.62.

Muhammad Abdul-Mageed, Amr Keleg, Abdel-Rahim Elmadany, Chiyu Zhang, Injy Hamed, Walid Magdy, Houda Bouamor, and Nizar Habash. NADI 2024: The fifth nuanced Arabic dialect identification shared task. In Nizar Habash, Houda Bouamor, Ramy Eskander, Nadi Tomeh, Ibrahim Abu Farha, Ahmed Abdelali, Samia Touileb, Injy Hamed, Yaser Onaizan, Bashar Alhafni, Wissam Antoun, Salam Khalifa, Hatem Haddad, Imed Zitouni, Badr AlKhamissi, Rawan Almatham, and Khalil Mrini, editors, *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 709–728, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.arabicnlp-1.79. URL https://aclanthology.org/2024.arabicnlp-1.79/.

Arun Babu, Changhan Wang, Andros Tjandra, Kushal Lakhotia, Qiantong Xu, Naman Goyal, Kritika Singh, Patrick von Platen, Yatharth Saraf, Juan Pino, Alexei Baevski, Alexis Conneau, and Michael Auli. XLS-R: Self-supervised Cross-lingual Speech Representation Learning at Scale. In *Interspeech 2022*, pages 2278–2282, 2022. doi: 10.21437/Interspeech.2022-143.

Emanuele Bastianelli, Andrea Vanzo, Pawel Swietojanski, and Verena Rieser. SLURP: A spoken language understanding resource package. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7252–7262. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.emnlp-main.588.

Haroun Elleuch, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. SLURP-TN: Resource for tunisian dialect spoken language understanding. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 1544–1552, Palma de Mallorca, Spain, 2026. ELRA Language Resource Association. doi: 10.63317/4m2ac973aco4.

Stefan Larson, Anish Mahendran, Joseph J. Peper, Christopher Clarke, Andrew Lee, Parker Hill, Jonathan K. Kummerfeld, Kevin Leach, Michael A. Laurenzano, Lingjia Tang, and Jason Mars. An evaluation dataset for intent classification and out-of-scope prediction. In *Proceedings of EMNLP-IJCNLP*, pages 1311–1316. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1131.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12(85):2825–2830, 2011.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya

Sutskever. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning*, pages 28492–28518. PMLR, 2023.

Yilin Shen, Yen-Chang Hsu, Avik Ray, and Hongxia Jin. Enhancing the generalization for intent classification and out-of-domain detection in SLU. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*, pages 2443–2453. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.acl-long.190.

Peter Sullivan, Bashar Talafha, Ahmed Ashraf, Fethi Bougares, Haroun Elleuch, Chiyu Zhang, AbdelRahim Elmadany, Youssef Mohamed, Salima Mdhaffar, Yannick Estève, Mohamed Elhoseiny, Hamzah Luqman, Nizar Habash, and Muhammad Abdul-Mageed. NADI-2026: The Second Multidialectal Arabic Speech Processing Shared Task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary, 2026. Association for Computational Linguistics.

Bashar Talafha, Hawau Olamide Toyin, Peter Sullivan, AbdelRahim A. Elmadany, Abdurrahman Juma, Amirbek Djanibekov, Chiyu Zhang, Hamad Alshehhi, Hanan Aldarmaki, Mustafa Jarrar, Nizar Habash, and Muhammad Abdul-Mageed. NADI 2025: The first multidialectal Arabic speech processing shared task. In Kareem Darwish, Ahmed Ali, Ibrahim Abu Farha, Samia Touileb, Imed Zitouni, Ahmed Abdelali, Sharefah Al-Ghamdi, Sakhar Alkhereyf, Wajdi Zaghouani, Salam Khalifa, Badr AlKhamissi, Rawan Almatham, Injy Hamed, Zaid Alyafeai, Areeb Alowisheq, Go Inoue, Khalil Mrini, and Waad Alshammari, editors, *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 720–733, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-356-2. doi: 10.18653/v1/2025.arabicnlp-sharedtasks.99. URL https://aclanthology.org/2025.arabicnlp-sharedtasks.99/.